ECE 3355: Electronics

Section 23780/29962
Spring 2021

Exam 2

Version A

April 22, 2021

Complete the exam on your own, without the help of your notes, prior examples or solutions, your book, or any communication/interaction with others. You must write a complete solution that shows the relevant steps if you want full credit for the problem. You may use a calculator and a crib sheet is provided as part of this exam. **You will have 1 hr. 45 min. to finish the exam.**

Student's Name: Solution

Question	Points	Score
1	30	
2	30	
3	40	
Total:	100	

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 7.3 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

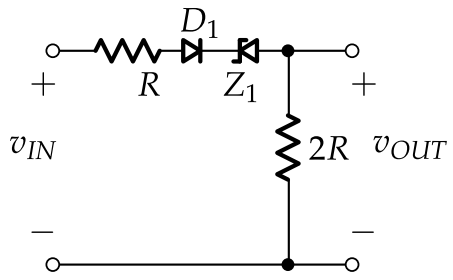
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. (30 points) For the circuit below, assume $V_{D0} = 0.7$ volts (for both diodes), $V_{Z0} = 1.8$ volts, and $r_D = r_Z = 0$.

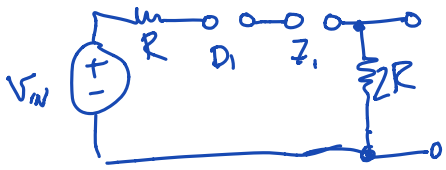


(a) Write an expression for v_{OUT} in terms of v_{IN}.

Two possibilities:

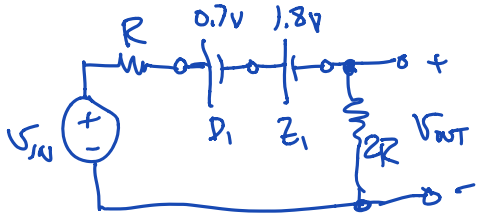
(1) Both diodes are "off"

Note: Z_1 cannot be forward biased since it is in series with D_1



$V_{OUT} = 0$

(2) D_1 is forward biased and Z_1 is in reverse breakdown:

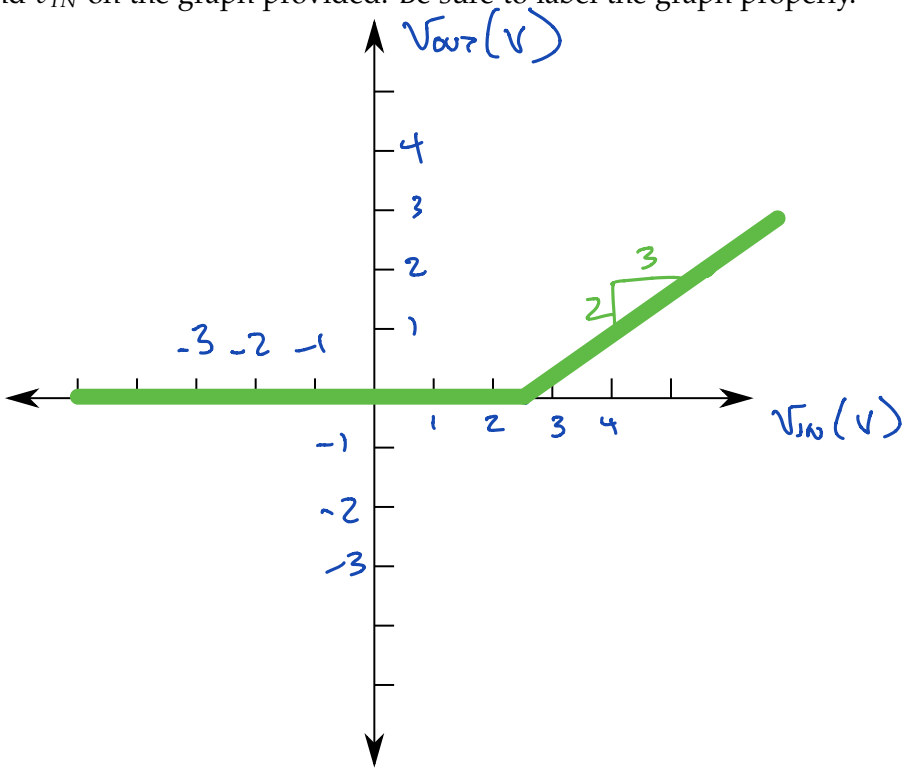


$V_{OUT} = (V_{IN} - 2.5V) \cdot \frac{2R}{3R} = \frac{2}{3}(V_{IN} - 2.5V)$

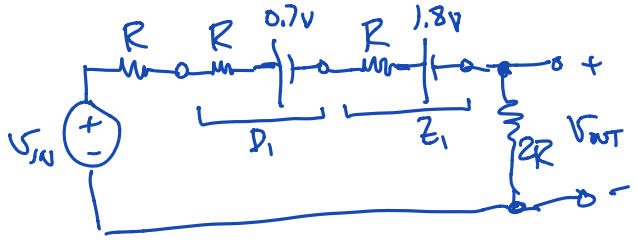
Hence

$$V_{OUT} = \begin{cases} 0 & V_{IN} \le 2.5V \\ \frac{2}{3}(V_{IN} - 2.5V) & V_{IN} > 2.5V \end{cases}$$

(b) Sketch v_{OUT} and v_{IN} on the graph provided. Be sure to label the graph properly.



(c) Write an expression for v_{OUT} in terms of v_{IN} if $r_Z = r_D = R$.

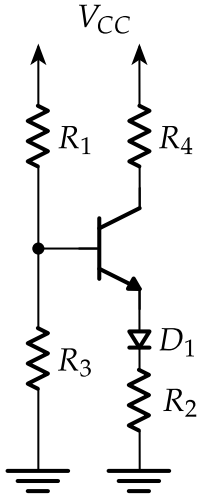


$$v_{OUT} = (v_{IN} - 2.5V)\frac{2R}{5R}$$

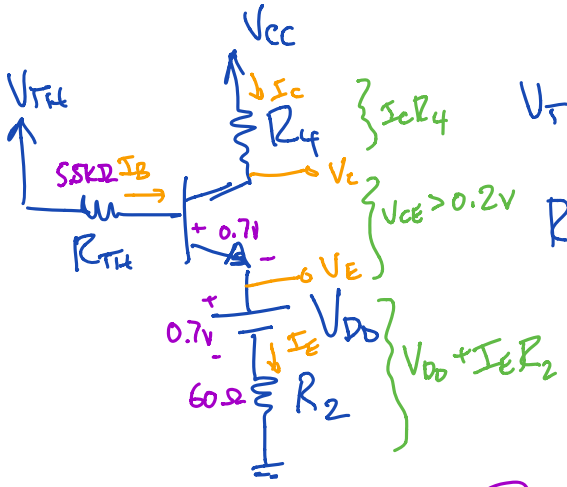
So:

$$v_{OUT} = \begin{cases} 0 & v_{IN} \leq 2.5V \\ \frac{2}{5}(v_{IN} - 2.5V) & v_{IN} > 2.5V \end{cases}$$

2. (30 points) For the following circuit, find the range of values for R_4 that will ensure that the circuit remains in the active mode if $\beta = 99$. You may use the constant voltage drop model for the diode with $V_{D0} = 0.7$ volts. Also, $R_1 = R_3 = 11\,\text{k}\Omega$, $R_2 = 60\,\Omega$, and $V_{CC} = 12\,\text{V}$.



If transistor is active, then D_1 will be forward biased ← our initial guess



$$V_{TH} = \frac{R_3}{R_1 + R_3} \cdot V_{CC} = 6\,\text{V}$$

$$R_{TH} = R_1 \| R_3 = 5.5\,\text{k}\Omega$$

$$V_{TH} = I_B R_{TH} + 0.7\text{V} + 0.7\text{V} + I_E R_2 \qquad I_E = (\beta+1) I_B$$

$$I_B = \frac{V_{TH} - 1.4\text{V}}{R_{TH} + (\beta+1) R_2} = 0.4\,\text{mA}$$

$$I_E = 40\,\text{mA}$$

$$I_C = \beta I_B = 39.6\,\text{mA}$$

$$V_E = I_E R_2 + V_{DO} = 3.1\,V$$

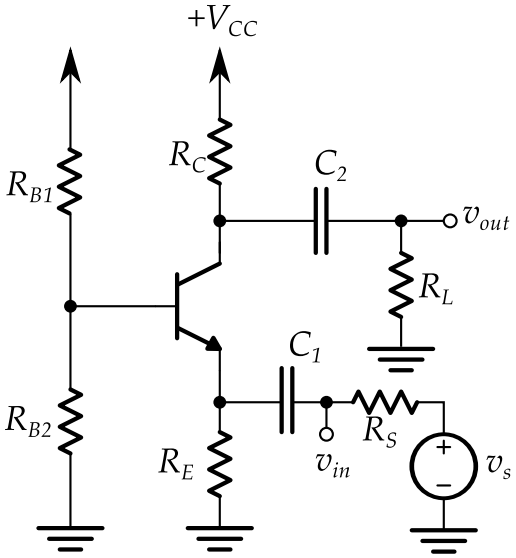
for the transistor to be active (it is not in cut-off), $V_{CE} > 0.2\,V$

$$V_{CE} = V_C - V_E = V_{CC} - I_C R_4 - V_E > 0.2\,V$$

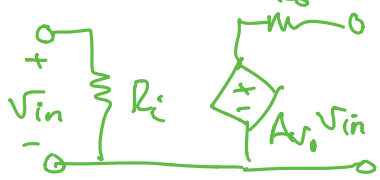
Hence,

$$R_4 < \frac{V_{CC} - V_E - 0.2V}{I_C} = 220\,\Omega$$

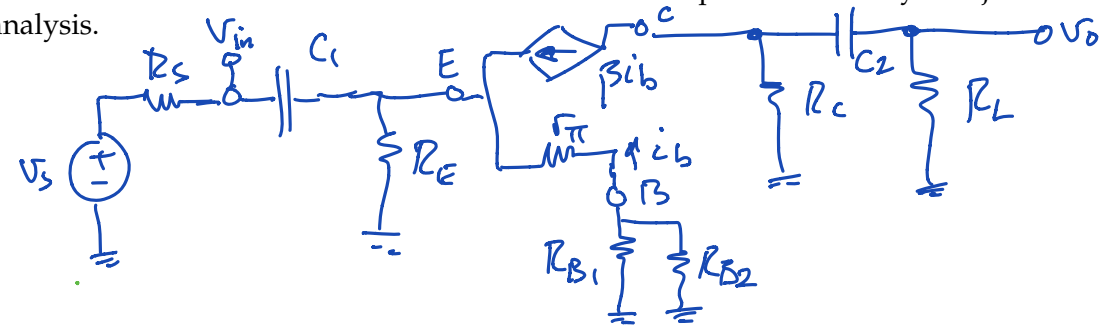
3. Use the following circuit to complete this problem.



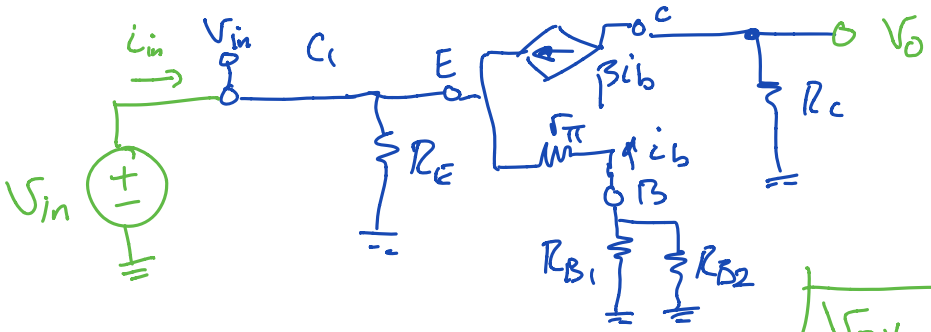
Goal: find parameters of simple amp model



(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit. *Hint:* You do not need to complete a DC analysis - just the AC analysis.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

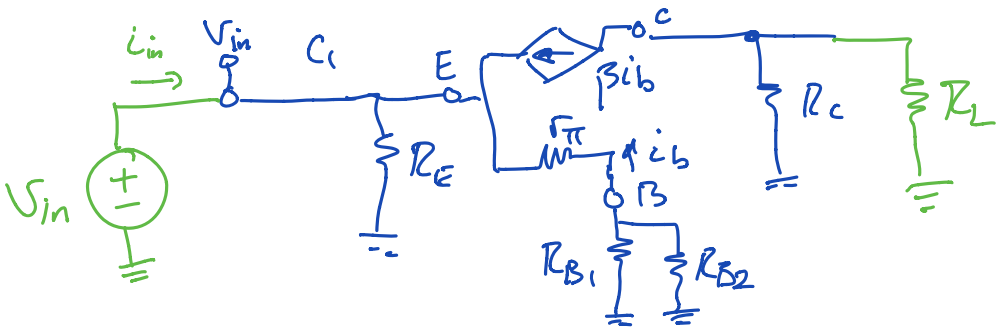


$V_o = -\beta i_b R_C$

$V_{in} = -i_b(r_\pi + R_{B1} \| R_{B2})$

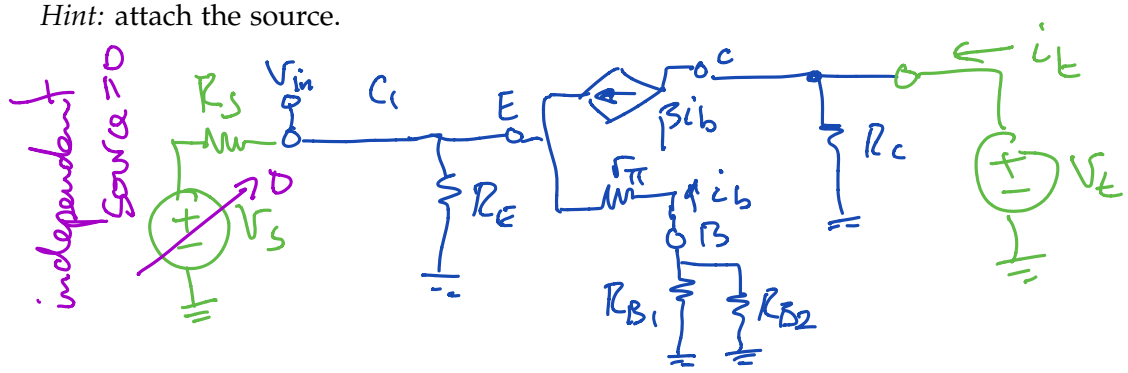
$$\frac{V_{out}}{V_{in}} = \frac{\beta R_C}{r_\pi + R_{B1} \| R_{B2}}$$

(c) (10 points) Find expression for the input resistance for the *midband*.
Hint: attach the load.



$$R_{in} = R_E \| \left[\frac{v_{in}}{(\beta+1) i_b} \right] = R_E \| \left[\frac{-i_b (r_\pi + R_{B1} \| R_{B2})}{-(\beta+1) i_b} \right] = R_E \| \left[\frac{r_\pi + R_{B1} \| R_{B2}}{\beta+1} \right]$$

(d) (10 points) Find expression for the output resistance for the *midband*.
Hint: attach the source.



Note: $i_b (R_{B1} \| R_{B2}) + i_b r_\pi + (\beta+1) i_b \cdot (R_E \| R_S) = 0$

So: $i_b = 0$

$\beta i_b = 0$

$$R_o = \frac{V_t}{i_t} = \frac{i_t \cdot R_C}{i_t} = \boxed{R_C}$$